



Year 12 Mathematics Extension 1

2019

Assessment Task 1

Full Name

Please circle your teacher's name

Samuel

Siriwardena

General Instructions

- Reading Time – 5 minutes
- Working Time – 55 minutes
- Write using a black pen
- Do not use liquid paper or tape
- Approved calculators may be used
- You may use the reference sheet provided during the examination.
- Diagrams are not to scale

Total marks – 30

Attempt both Questions.

Answer the questions in the space provided.
These spaces provide guidance for the expected length of response.

Your responses should include relevant mathematical reasoning/ or calculations.

Question No.	1	2	Total
Marks	/9	/21	/30

Answer each question in the space provided.

QUESTION 1 (12 marks)

Marks

a) Use mathematical induction to prove that for all positive integers of n:

3

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n + 1)!$$

[illegible]

b) Use mathematical induction to prove that for all positive integers of n:

3

$5^n + 2 \times 11^n$ is divisible by 3.

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QUESTION 2 (21 marks)

a) (i) Express $\cos t - \sqrt{3} \sin t$ in the form $R \cos(t + \alpha)$, where α is in radians.

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(ii) Hence or otherwise, find the all the solutions to the equation

$$\cos \theta - \sqrt{3} \sin \theta = \sqrt{2} \text{ for } 0 \leq \theta \leq 2\pi.$$

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3QUESTION 2

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b) Find all the solutions to the equation

$$\cos \theta = \sin 2\theta, \text{ for } 0 \leq \theta \leq 2\pi.$$

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c) Find all the solutions to the equation

$$\sin 3x + \sin x = \sin 2x, \text{ for } 0 \leq x \leq 2\pi.$$

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3 i) Find all

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3 j) Find all

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d) Solve the following equation using t formulae for $0 \leq x \leq 360^\circ$.

$$5 \cos x + 3 \sin x = 4$$

3 k) Solve the

e) Prove that $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$ for $\sin \theta \neq 0$ and $\cos \theta \neq 0$.

2 1) Prove th

If you use this

[illegible]

Marks

3

3



Year 12 Mathematics Extension 1

2019

Assessment Task 1

Full Name *Solutions*

Please circle your teacher's name

Samuel

Siriwardena

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Question No.	1	2	Total
Marks	/9	/21	/30

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QUESTION 1 (12 marks)

Marks

a) Use mathematical induction to prove that for all positive integers of n :

3

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$$

$$n=1, \dots \text{LHS} = 2 \times 1! = 2, \dots \text{RHS} = 1 \times 2! = 2$$

$\therefore$ The result is true for $n=1$

Assume the result is true for $n=k$.

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! = k(k+1)! \rightarrow \textcircled{A}$$

When $n=k+1$, the result to be proved,

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! + [(k+1)^2 + 1](k+1)! = (k+1)(k+2)!$$

$$\text{LHS} = \dots \text{from } \textcircled{A}$$

$$= k(k+1)! + [(k+1)^2 + 1](k+1)!$$

$$= (k+1)! \{ k + (k+1)^2 + 1 \}$$

$$= (k+1)! \{ k + k^2 + 2k + 1 + 1 \}$$

$$= (k+1)! (k^2 + 3k + 2)$$

$$= (k+1)! (k+2)(k+1)$$

$$= (k+1)(k+2)! = \text{RHS}$$

$\therefore$ The result is true for $n=k+1$.

$\therefore$ By mathematical induction, the result is true for all positive integers of n .

b) Use mathematical induction to prove that for all positive integers of n :

3

$5^n + 2 \times 11^n$ is divisible by 3.

When $n=1$, $5^1 + 2 \times 11^1 = 27$ is divisible by 3.

Assume the result is true for $n=k$.

$$\therefore 5^k + 2 \times 11^k = 3M \text{ where } M \text{ is an integer.}$$

$$5^k = 3M - 2 \times 11^k \rightarrow \textcircled{A}$$

When $n=k+1$,

$$5^{k+1} + 2 \times 11^{k+1}$$

$$= 5 \times 5^k + 2 \times 11 \times 11^k$$

$$= 5(3M - 2 \times 11^k) + 22 \times 11^k$$

$$= 15M - 10 \times 11^k + 22 \times 11^k$$

$$= 15M + 12 \times 11^k$$

$$= 3(5M + 4 \times 11^k)$$

is divisible by 3

$\therefore$ The result is true for $n=k+1$.

$\therefore$ By mathematical induction, the result is true for all positive integers of n .

c) (i) Suppose that the statement

$1 + 3 + 5 + \dots + (2n - 1) = n^2 + 2$ is true for the positive integer

$n = k$. Prove it is also true for $n = k + 1$.

$$\begin{aligned}
 & 1 + 3 + 5 + \dots + (2n - 1) = n^2 + 2 \\
 \text{when } n = k, & 1 + 3 + 5 + \dots + (2k - 1) = k^2 + 2 \\
 \text{when } n = k + 1, & \text{LHS} = 1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) \\
 & = k^2 + 2 + 2k + 1 \\
 & = k^2 + 2k + 3 \\
 & = k^2 + 2k + 1 + 2 \\
 & = (k + 1)^2 + 2 \\
 \text{when } n = k + 1, & \text{RHS} = (k + 1)^2 + 2 \\
 & \therefore \text{LHS} = \text{RHS} \\
 & \therefore \text{The result is true for } n = k + 1.
 \end{aligned}$$

(ii) Explain, by giving mathematical reasoning, why we cannot conclude that the

statement is true for all integers $n \geq 1$.

$$\begin{aligned}
 & \text{when } n = 1, \text{LHS} = 1, \text{RHS} = 1^2 + 2 = 3 \\
 & \therefore \text{LHS} \neq \text{RHS} \\
 & \therefore \text{The result is not true for } n = 1 \\
 & \therefore \text{we can't conclude that the statement is true for all integers } n \geq 1
 \end{aligned}$$

End of Question 1

QUESTION 2 (21 marks)

- a) (i) Express $\cos t - \sqrt{3} \sin t$ in the form $R \cos(t + \alpha)$, where α is in radians. 3

$$R \cos(t + \alpha) = R (\cos t \cos \alpha - \sin t \sin \alpha) = \cos t - \sqrt{3} \sin t$$

By equating the coefficients of $\cos t$ and $\sin t$ on both sides,

$$R \cos \alpha = 1 \rightarrow \textcircled{1}$$

$$R \sin \alpha = \sqrt{3} \rightarrow \textcircled{2}$$

$$\textcircled{2}/\textcircled{1} \quad \tan \alpha = \sqrt{3} \quad \therefore \alpha = \frac{\pi}{3}$$

$$\textcircled{1}^2 + \textcircled{2}^2 \quad R^2 (\cos^2 \alpha + \sin^2 \alpha) = 4 \quad \therefore R = 2$$

$$\therefore \cos t - \sqrt{3} \sin t = 2 \cos(t + \frac{\pi}{3})$$

- (ii) Hence or otherwise, find all the solutions to the equation

2

$$\cos \theta - \sqrt{3} \sin \theta = \sqrt{2} \text{ for } 0 \leq \theta \leq 2\pi.$$

$$\cos \theta - \sqrt{3} \sin \theta = 2 \cos(\theta + \frac{\pi}{3}) \text{ (from part (i))}$$

$$\therefore 2 \cos(\theta + \frac{\pi}{3}) = \sqrt{2}$$

$$0 \leq \theta \leq 2\pi$$

$$\cos(\theta + \frac{\pi}{3}) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} = \cos \frac{\pi}{4} \quad \frac{\pi}{3} \leq \theta + \frac{\pi}{3} \leq 2\pi + \frac{\pi}{3}$$

$$\therefore \theta + \frac{\pi}{3} = \frac{\pi}{4}, \text{ or } \theta + \frac{\pi}{3} = 2\pi + \frac{\pi}{4}, \text{ or } \theta + \frac{\pi}{3} = 2\pi + \frac{\pi}{4}$$

$$\theta = \frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$$

$$\theta = 2\pi + \frac{\pi}{4} - \frac{\pi}{3} = \frac{17\pi}{12}$$

$$\theta = 2\pi + \frac{\pi}{4} - \frac{\pi}{3} = \frac{23\pi}{12}$$

Not in the given domain

$$\therefore \text{Solutions are } \theta = \frac{17\pi}{12} \text{ or } \frac{23\pi}{12}$$

b) Find all the solutions to the equation

3

$$\cos \theta = \sin 2\theta, \text{ for } 0 \leq \theta \leq 2\pi.$$

$$\sin 2\theta - \cos \theta = 0$$

$$2\sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta (2\sin \theta - 1) = 0$$

$$\cos \theta = 0 \text{ or } \sin \theta = \frac{1}{2} = \sin \frac{\pi}{6}$$

$$\therefore \theta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \quad \theta = \frac{\pi}{6} \text{ or } \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\therefore \text{Solutions are } \theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6} \text{ or } \frac{3\pi}{2}$$

c) Find all the solutions to the equation

3

$$\sin 3x + \sin x = \sin 2x, \text{ for } 0 \leq x \leq 2\pi.$$

$$\sin 3x + \sin x = 2\sin 2x \cos x \quad (\text{using } A+B \text{ formulae})$$

$$\therefore 2\sin 2x \cos x = \sin 2x$$

$$\therefore 2\sin 2x \cos x - \sin 2x = 0$$

$$\sin 2x (2\cos x - 1) = 0$$

$$\therefore \sin 2x = 0 \quad \text{or} \quad 2\cos x - 1 = 0$$

$$\therefore 2x = 0, \pi \text{ or } 2\pi$$

$$\cos x = \frac{1}{2} = \cos \frac{\pi}{3}$$

$$x = 0, \frac{\pi}{2} \text{ or } \pi$$

$$\therefore x = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

$$\therefore \text{Solutions are } 0, \frac{\pi}{3}, \frac{\pi}{2}, \pi \text{ or } \frac{5\pi}{3}$$

d) Solve the following equation using t formulae for $0 \leq x \leq 360^\circ$.

3

$$5 \cos x + 3 \sin x = 4$$

$$5 \left(\frac{1-t^2}{1+t^2} \right) + 3 \times \frac{2t}{1+t^2} = 4$$

$$\frac{5-5t^2}{1+t^2} + \frac{6t}{1+t^2} = 4(1+t^2)$$

$$9t^2 - 6t - 1 = 0$$

$$(3t-1)^2 - 2 = 0$$

$$(3t-1)^2 = 2$$

$$3t-1 = \pm\sqrt{2}$$

$$t = \frac{1 \pm \sqrt{2}}{3}$$

$$\tan \frac{x}{2} = \frac{1+\sqrt{2}}{3}, \quad \text{or} \quad \tan \frac{x}{2} = \frac{1-\sqrt{2}}{3} = \tan(-7^\circ 52')$$

$$\tan \frac{x}{2} = \tan 38^\circ 49' \quad \text{or} \quad \frac{x}{2} = 180^\circ - 7^\circ 52' \text{ or}$$

$$\tan \frac{x}{2} = \tan(180^\circ + 38^\circ 49') \quad \frac{x}{2} = 360^\circ - 7^\circ 52'$$

$$\frac{x}{2} = 38^\circ 49' \text{ or } \frac{x}{2} = 218^\circ 49'$$

$$x = 2(180^\circ - 7^\circ 52')$$

$$\frac{x}{2} = 77^\circ 38'$$

$$x = 344^\circ 16'$$

outside the domain

outside the domain

$\therefore$ Solutions are $x = 77^\circ 38'$ or $344^\circ 16'$

e) Prove that $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$ for $\sin \theta \neq 0$ and $\cos \theta \neq 0$.

2

$$\text{LHS} = \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin \theta \cos \theta}$$

$$= \frac{\sin(3\theta - \theta)}{2 \sin \theta \cos \theta}$$

$$= \frac{2 \sin 2\theta}{2 \sin \theta \cos \theta}$$

$$= \frac{2 \sin 2\theta}{\sin 2\theta}$$

$$= 2 = \text{RHS}$$

f) Solve the equation $\sin 2x = 2\sin^2 x$, for $0 \leq x \leq \pi$.

3

$$\sin 2x - 2\sin^2 x = 0$$

$$2\sin x \cos x - 2\sin^2 x = 0$$

$$2\sin x (\cos x - \sin x) = 0$$

$$\sin x = 0 \quad \text{or} \quad \cos x - \sin x = 0$$

$$x = 0, \pi, \pi$$

$$\tan x = 1 = \tan \frac{\pi}{4}$$

$$x = \frac{\pi}{4} \text{ or } x = \frac{\pi}{4} + \pi \leftarrow \text{outside the domain.}$$

$\therefore$ Solutions are $0, \frac{\pi}{4}, \pi$.

g) Using the formula, $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$, simplify:

2

$$\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} - 1 \text{ for } 0 < \theta < \frac{\pi}{2}.$$

$$\frac{(\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)}{(\sin \theta + \cos \theta)} - 1$$

$$= 1 - \sin \theta \cos \theta - 1$$

$$= -\sin \theta \cos \theta.$$

End of paper 😊

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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